

DEVSECOPS & PLATFORM · ISSUE 04 OF 08

Helm vs Kustomize.

Pick by who owns the YAML, not by which tool is trendier.

04

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Swipe →

Different jobs

HELM

- Templating + packaging
- Values files
- Release history + rollback
- Great for third-party apps

KUSTOMIZE

- Patches on plain YAML
- Overlays per environment
- No template language
- Great for your own services



Which one when

SITUATION	CHOICE
Installing third-party software	Helm chart
Own service, small env differences	Kustomize
Many near-identical services	Helm library chart
Vendor chart needs a tweak	Helm + Kustomize patch



A small overlay

YAML

```
apiVersion: kustomize.config.k8s.io/v1beta1
kind: Kustomization
resources:
  - ../../base
patches:
  - path: replicas.yaml
images:
  - name: api
    newTag: "1.4.2"
```



Keep it reviewable

DO

- ✓ Render and diff in CI
- ✓ Validate output with kubeconform
- ✓ Review rendered YAML, not templates only

AVOID

- ✗ Charts with 40 if-blocks
- ✗ Overlays copying whole files
- ✗ Logic hidden in helpers nobody reads

Render before you merge

BASH

```
helm template api ./chart -f values-prod.yaml \  
| kubeconform -strict -summary  
  
kustomize build apps/overlays/prod \  
| kubeconform -strict -summary
```



KEY TAKEAWAY

Helm for packages, Kustomize for overlays, CI renders both.

- Third-party: Helm
- Own services: Kustomize
- Always review rendered output

YOUR TURN

Which do you use for your own services?

FOLLOW AKHILESH RANJAN SINGH

Next in the series: #05 Terraform at scale

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